PHYS 4041/5041:

Mathematical Methods of Physics 1

PDEs in Spherical Coordinates,

Legendre Polynomials,

and Related Topics

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Separation of the Laplacian in spherical coordinates

- See Boas, section 13.7.
- We will consider the PDE $(\nabla^2 + k^2)\Psi = 0$ in spherical coordinates. This is the Helmholtz equation. Many important cases are included in this example, or are very similar.
- Recall, that in spherical coordinates,

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\,.$$

- Assume $\Psi = R(r)\,\Theta(\theta)\,\Phi(\phi)$, then

$$\frac{1}{r^2 R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{1}{r^2\sin\theta}\frac{1}{\Theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} = -k^2.$$

Separation of the Laplacian in spherical coordinates

- Multiple through by $r^2 \sin^2\theta$. This isolates the ϕ-dependent part. It must be constant:

$$\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} = -m^2.$$

- We have chosen the separation constant to be negative. This is because, that in most physical applications, we want $\Phi(0) = \Phi(2\pi)$. We therefore demand that $\Phi(\phi)$ have a period of 2π. This cannot be achieved with a positive separation constant.

- We know how to solve this equation. The solutions are $\cos(m\phi)$, $\sin(m\phi)$, or (including both of the previous) $\exp(im\phi)$, where m is an integer.

Azimuthal symmetry

- For now, let's assume that our problem is azimuthally symmetric. This means we look for solutions with no ϕ dependence. In our present notation, this implies $\Phi = \text{constant}$ and $m = 0$.

- This results in

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + kr^2 + \frac{1}{\sin\theta}\frac{1}{\Theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) = 0.$$

- This equation is again separated. For the θ dependence, we can take

$$\frac{1}{\sin\theta}\frac{1}{\Theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) = -\lambda.$$

Azimuthal symmetry, continued

- Now, we charge variables to $x = \cos\theta$:

$$(1 - x^2)\frac{d^2\Theta}{dx^2} - 2x\frac{d\Theta}{dx} + \lambda\Theta = 0.$$

- If we set $\lambda = \ell(\ell + 1)$, we recognize this as the Legendre differential equation.

- The radial equation is

$$\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + [kr^2 - \ell(\ell + 1)]R = 0.$$

- If $k = 0$, the solutions are r^ℓ and $r^{-(\ell+1)}$. We will defer the discussion of the more general radial equation until later.

Legendre differential equation

- The Legendre differential equation is (see Boas, section 12.2):

$$(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + \ell(\ell+1)y = 0.$$

- There are regular singular points at $x = \pm 1$.

- We can use the Frobenius method for a series around $x = 0$:

$$y(x) = x^s \sum_{n=0}^{\infty} a_n x^n.$$

- The indicial equation is $s(s-1) = 0$. The Frobenius method yields

$$a_{n+2} = \frac{(n+s+\ell+1)(n+s-\ell)}{(n+s+1)(n+s+2)} a_n.$$

Legendre differential equation, continued

- For $s = 0$, we separate the solution into even and odd parts:

$$y = a_0(1 + \alpha_2 x^2 + \alpha_4 x^4 + \ldots) + a_1 x(1 + \beta_3 x^2 + \beta_5 x^4 + \ldots),$$

  where a_0 and a_1 are arbitrary constants and the α_n and β_n are defined by the recurrence formula.

- The $s = 1$ case is redundant with the odd part above, so it need not be considered further.

- Here we assume the boundary condition that $y(x)$ must be finite for $x \to \pm 1$. We also observe that $\lim_{n\to\infty} a_{n+2}/a_n = 1$.

- The *only* way to get finite solutions at $x = \pm 1$ is if the series terminates.

Legendre differential equation, continued

- From the recurrence formula, we observe that a_{n+2} (and hence all subsequent terms) will vanish if $n = -(\ell + 1)$ or $n = \ell$.
- We thus require that ℓ is *an integer*.
- We observe that negative integer values for ℓ are redundant, since $\ell(\ell + 1) = [-(\ell + 1)][-(\ell + 1) + 1]$. We thus require $\ell = 0, 1, 2, \ldots$.
- Furthermore, we see that we cannot have both a_0 and a_1 non-zero, since we cannot make both series terminate for a single ℓ value.
- Therefore, we take $a_1 = 0$ when ℓ is even and $a_0 = 0$ when ℓ is odd and the solution is then an even or odd function, depending on whether ℓ is even or odd.

Legendre differential equation, continued

- The Frobenius method and boundary conditions have restricted ℓ to be a non-negative integer and defined the solutions, up to an arbitrary normalization constant.
- We now define our solutions to be the *Legendre polynomials* $P_\ell(x)$, which are normalized such that $P_\ell(1) = 1$. This is simply a convention that leads to the simplest formulas in what follows.
- The first few are:

$$P_0 = 1, \qquad P_1 = x, \qquad P_2 = \frac{3}{2}x^2 - \frac{1}{2}.$$

- Note that $P_\ell(1) = 1$ implies $P_\ell(-1) = (-1)^\ell$. Why?

Sturm-Liouville connections

- You have probably noticed that the Legendre ODE together with the boundary conditions of finite solutions $x = \pm 1$ define a Sturm-Liouville eigenvalue problem.
- We can conclude immediately that the Legendre polynomials P_ℓ, with $\ell = 0, 1, 2, \ldots$ form a complete basis over the interval $[-1, 1]$ and are also orthogonal.
- The orthogonality relation (the weight function is unity and we will derive this normalization factor later) is

$$\int_{-1}^{1} P_m(x)\, P_n(x)\, dx = \frac{2}{2m+1}\delta_{mn}.$$

- Note that the functions are *not* normalized. This is a consequence of choosing $P_m(1) = 1$.
- The orthogonality and completeness are crucial for matching boundary conditions in PDEs.

Legendre polynomials, continued

- Any square-integrable $f(x)$ defined over $-1 \le x \le 1$ can thus be represented

$$f(x) = \sum_{\ell=0}^{\infty} A_\ell \, P_\ell(x)$$

$$A_\ell = \frac{2\ell+1}{2} \int_{-1}^{1} f(x) \, P_\ell(x) \, dx.$$

- If you require *orthonormal* functions, you can always define $u_\ell(x) = \sqrt{\frac{2\ell+1}{2}} \, P_\ell(x)$.

- We also have the completeness relation for x,x' in $(-1, 1)$:

$$\sum_{\ell=0}^{\infty} \frac{2\ell+1}{2} \, P_\ell(x') \, P_\ell(x) = \delta(x - x').$$

Legendre polynomials, continued

- If $f(x)$ is a polynomial of degree ℓ, there is another way. Note there are two bases that can be used:

$$\{1, x, \ldots, x^\ell\}$$
$$\{P_0, P_1, \ldots, P_\ell\}$$

  They are both linearly independent and complete over the space of polynomials of degree $\leq \ell$.

- In this case, it is often easier to determine the A_ℓ using linear algebra techniques, as opposed to integration.

- Note that P_ℓ is the only Legendre polynomial of degree $\leq \ell$ that includes x^ℓ (i.e., ℓ is the *degree* of the Legendre polynomial P_ℓ). Start here!

Laplace equation with azimuthal symmetry

- Let's go back so solving $\nabla^2\Psi = 0$ (i.e., $k = 0$) with $\Psi(r,\theta) = R(r)\,\Theta(\theta)$. The radial equation is

$$\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) - \ell(\ell+1)R = 0.$$

- The solutions for $R(r)$ are r^ℓ and $r^{-(\ell+1)}$.
- If we are solving $\nabla^2\Psi = 0$ for an azimuthally-symmetric problem that includes $0 \le \theta \le \pi$, the solution must then be

$$\Psi(r,\theta) = \sum_{\ell=0}^{\infty}\left(A_\ell r^\ell + \frac{B_\ell}{r^{\ell+1}}\right)P_\ell(\cos\theta),$$

  where A_ℓ and B_ℓ are constants determined by boundary conditions.
- If our region includes the origin (or $r \to \infty$) and Ψ is required to be finite there, then we must have $B_\ell = 0$ (or $A_\ell = 0$).

Generating function for the Legendre polynomials

- Consider the function

$$\Psi(r,\theta) = \frac{1}{\sqrt{a^2 - 2ar\cos\theta + r^2}} = \frac{1}{a}\frac{1}{\sqrt{1 - 2hx + h^2}},$$

  where $a > 0$ is a constant, $h = r/a$, and $x = \cos\theta$.

- One can show in a few lines of algebra that $\nabla^2\Psi = 0$. Physically, this Ψ could represent the potential of a point charge located on the $+z$ axis at $(0, 0, a)$ – remember the law of cosines!

- Since Ψ is clearly finite at $r = 0$, we can expand it as

$$\Psi(r,\theta) = \sum_{\ell=0}^{\infty} A_\ell\, r^\ell\, P_\ell(x) = \sum_{\ell=0}^{\infty} \alpha_\ell\, h^\ell\, P_\ell(x).$$

  We assume here that $h < 1$ (i.e., $r < a$).

Generating function, continued

- At $\theta = 0$ ($x = 1$), we have

$$\Psi(r,0) = \frac{1}{a}\frac{1}{1-h} = \frac{1}{a}\left(1 + h + h^2 + \ldots\right)$$
$$= \sum_{\ell=0}^{\infty} \alpha_\ell \, h^\ell \, P_\ell(1) = \sum_{\ell=0}^{\infty} \alpha_\ell \, h^\ell .$$

- Since the power series in h must be unique, we can conclude that $\alpha_\ell = 1/a$.

- Thus have

$$\Phi(x,h) = \frac{1}{\sqrt{1 - 2xh + h^2}} = \sum_{\ell=0}^{\infty} h^\ell \, P_\ell(x),$$

where $\Phi(x,h)$ is called the *generating function* for the Legendre polynomials.

Generating function, continued

- In addition to providing the solution to the Poisson equation for a point charge located away from the origin, the generating function is useful for deriving properties of the Legendre polynomials.
- The usual strategy is to first derive an identity involving the closed form $\Phi(x,h) = (1 - 2xh + h^2)^{-1/2}$. Then see what it implies for the Legendre polynomials, using the uniqueness of the power series in h.
- This reasoning typically yields an relation that holds for arbitrary ℓ.

Generating function → recurrence formula

- With a little algebra, one can show

$$(1 - 2xh + h^2)\frac{\partial \Phi}{\partial h} = (x - h)\Phi.$$

- This implies

$$\begin{aligned} 0 &= \sum_{\ell=0}^{\infty} \left(\ell h^{\ell-1} - 2\ell x h^{\ell} + \ell h^{\ell+1} - x h^{\ell} + h^{\ell+1}\right) P_\ell \\ &= P_1 - xP_0 + \sum_{\ell=2}^{\infty} \left[\ell P_\ell - (2\ell - 1)xP_{\ell-1} + (\ell - 1)P_{\ell-2}\right] h^{\ell-1}. \end{aligned}$$

- We conclude $P_1 = xP_0$ (which we already knew) and for $\ell \geq 2$

$$\ell P_\ell - (2\ell - 1)xP_{\ell-1} + (\ell - 1)P_{\ell-2} = 0.$$

Generating function → normalization factor

- Integrating the square of the generating function,

$$\int_{-1}^{1} \Phi^2 \, dx = \int_{-1}^{1} \frac{dx}{1 - 2xh + h^2} = \frac{1}{h} \log \frac{1+h}{1-h} = 2 \sum_{n=0}^{\infty} \frac{h^{2n}}{2n+1}.$$

- But we also have, using orthogonality,

$$\begin{aligned} \int_{-1}^{1} \Phi^2 \, dx &= \int_{-1}^{1} dx \sum_{m=0}^{\infty} h^m P_m \sum_{n=0}^{\infty} h^n P_n \\ &= \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} h^{m+n} \delta_{mn} \int_{-1}^{1} P_n^2 \, dx = \sum_{n=0}^{\infty} h^{2n} \int_{-1}^{1} P_n^2 \, dx. \end{aligned}$$

- We thus have

$$\int_{-1}^{1} P_n^2 \, dx = \frac{2}{2n+1}.$$

- This technique can also be used to prove orthogonality, but showing $\int_{-1}^{1} \Phi(x,h) \, \Phi(x,g) \, dx = \frac{1}{\sqrt{gh}} \log \frac{1+\sqrt{gh}}{1-\sqrt{gh}}$ is difficult.

Generating function → cosine representation

- We note that

$$(1-2h\cos\theta+h^2)^{-1/2} = (1-he^{i\theta})^{-1/2}(1-he^{-i\theta})^{-1/2}$$
$$= \left(1+\frac{1}{2}he^{i\theta}+\frac{3}{8}h^2e^{2i\theta}+\ldots\right)\left(1+\frac{1}{2}he^{-i\theta}+\frac{3}{8}h^2e^{-2i\theta}+\ldots\right).$$

- The resulting series can be arranged to show that

$$P_\ell(\cos\theta) = \sum_n^\ell a_{\ell n}\cos n\theta,$$

where n takes on the even (odd) non-negative integers $\leq \ell$ when ℓ is even (odd). Importantly, the $a_{\ell n}$ coefficients are *all positive*.

- Examples: $P_2(\cos\theta) = \frac{3}{4}\cos 2\theta + \frac{1}{4}$, $P_3(\cos\theta) = \frac{5}{8}\cos 3\theta + \frac{3}{8}\cos\theta$.

- We can also conclude $|P_\ell(\cos\theta)| \leq P_\ell(1) = 1$.

Rodrigues' formula

- Another interesting representation of the Legendre polynomials is

$$P_\ell(x) = \frac{1}{2^\ell \ell!} \left(\frac{d}{dx} \right)^\ell (x^2 - 1)^\ell.$$

- The derivation is discussed in Boas. Like the generating function, this formula is also useful for deriving properties of the Legendre polynomials.

- By expanding $(x^2 - 1)^\ell$ using the binomial formula, one finds

$$P_\ell(x) = \frac{1}{2^\ell} \sum_{k=0}^{[\ell/2]} (-1)^k \frac{(2\ell - 2k)!}{k!(\ell - k)!(\ell - 2k)!} x^{\ell - 2k}.$$

- Rodrigeues type formulas exist for a large class of solutions to Sturm-Liouville ODEs.

Second solution

- From the differential equation, we know the Wronskian must be $\propto (1-x^2)^{-1}$. A second solution is thus given by

$$y_2(x) = P_\ell(x) \int \frac{dx}{(1-x^2)P_\ell^2(x)},$$

  but it contains an unconstrained linear combination of $P_\ell(x)$.

- To be definitive, the second solution to the Legendre ODE is taken to be the part of the $s = 0$ Frobenius solution that we threw out before. In other words, the odd series when ℓ is even or the even series when ℓ is odd.

- These infinite series can be summed analytically. The formula for $y_2(x)$ above gives you a clue how to do it, especially when $\ell = 0$ or 1.

Second solution, continued

- The recipe above does not determine the absolute normalization.
- The first two second solutions are normalized in this way:

$$Q_0(x) = \frac{1}{2}\log\frac{1+x}{1-x} \quad \text{and} \quad Q_1(x) = \frac{x}{2}\log\frac{1+x}{1-x} - 1.$$

- The second solutions for higher ℓ are defined by assuming the $Q_\ell(x)$ satisfy the same recurrence formulas as the $P_\ell(x)$.
- These second solutions are knows as the Legendre functions of the second kind.
- All of them contain the $\log\frac{1+x}{1-x}$ factor that is singular at $x = \pm 1$.

Helmholtz equation without azimuthal symmetry

- The general result for $(\nabla^2 + k^2)[R(r)\,\Theta(\theta)\,\Phi(\phi)] = 0$, with $\Phi(\phi) = e^{im\phi}$ and m integral, is

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + kr^2 + \frac{1}{\sin\theta}\frac{1}{\Theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) - \frac{m^2}{\sin^2\theta} = 0.$$

- The separated equation for Θ, in terms of $x = \cos\theta$, is

$$(1-x^2)\frac{d^2\Theta}{dx^2} - 2x\frac{d\Theta}{dx} + \left[\ell(\ell+1) - \frac{m^2}{1-x^2}\right]\Theta = 0.$$

- This is known as the Associated Legendre ODE.

Associated Legendre functions

- We still require that $\Theta(x)$ be finite as $x \to \pm 1$, defining a Sturm-Liouville problem.
- One again finds that it is necessary for the Frobenius series to terminate, in order to have an acceptable solution. This now requires *two* restrictions: ℓ must be an integer – we take it to be non-negative for the same reasons as before – *and* $-\ell \leq m \leq \ell$.
- These solutions are known as the associated Legendre functions (see Boas section 12.10). For $m \geq 0$, they are defined to be

$$P_\ell^m(x) = (-1)^m (1-x^2)^{m/2} \left(\frac{d}{dx}\right)^m P_\ell(x).$$

- One can show fairly easily this is a solution to the ODE. Note also that $P_\ell^m(x)$ is *not* a polynomial in x when m is odd.

Associated Legendre functions, continued

- The definition can be extended to negative m using Rodrigues' formula

$$P_\ell^m(x) = \frac{(-1)^m}{2^\ell \ell!}(1-x^2)^{m/2}\left(\frac{d}{dx}\right)^{\ell+m}(x^2-1)^\ell.$$

  Note that some authors, such as Boas, do not include the factor of $(-1)^m$ in the definition. This factor once caused me to lose a day out of my life, that I will never get back.

- The orthogonality relation for a given m is

$$\int_{-1}^{1} P_\ell^m(x)\, P_{\ell'}^m(x)\, dx = \frac{2}{2\ell+1}\frac{(\ell+m)!}{(\ell-m)!}\,\delta_{\ell\ell'}.$$

Associated Legendre functions, continued

- Note that $P_\ell^{-m}(x) \propto P_\ell^m(x)$, i.e., they are not independent.
- Note that $P_\ell^0(x)$ is just $P_\ell(x)$.
- Note that
$$P_\ell^m(\pm 1) = \begin{cases} (\pm 1)^\ell & m = 0 \\ 0 & m \neq 0 \end{cases}.$$
- Some examples:

$$P_1^1(\cos\theta) = -\sin\theta, \quad P_2^1(\cos\theta) = -3\sin\theta\cos\theta, \quad P_2^2(\cos\theta) = 3\sin^2\theta.$$

- Note that $P_\ell^m(\cos\theta)$ *is* a bivariate polynomial of degree ℓ in $\cos\theta$ and $\sin\theta$.

Spherical harmonics

- In applications, the associated Legendre functions usually appear together with the angular functions $\exp(im\phi)$. Note their normalization and orthogonality:

$$\int_0^{2\pi} \left(e^{im\phi}\right)^* e^{im'\phi} d\phi = 2\pi\, \delta_{mm'}.$$

- It is thus useful to define the normalized spherical harmonics that are a function of θ *and* ϕ

$$Y_\ell^m(\theta,\phi) = \sqrt{\frac{2\ell+1}{4\pi}\frac{(\ell-m)!}{(\ell+m)!}}\, P_\ell^m(\cos\theta)\, e^{im\phi}.$$

- If we had not put in the factor of $(-1)^m$ before, we would have to put it in now. There is better agreement on the definition of the Y_ℓ^m. Our convention is called the Condon-Shortley phase convention.

Spherical harmonics, continued

- The spherical harmonics are orthonormal and complete:

$$\int_{-1}^{1} d\cos\theta \int_{0}^{2\pi} d\phi\, Y_\ell^{m*}(\theta,\phi)\, Y_{\ell'}^{m'}(\theta,\phi) = \delta_{\ell\ell'}\,\delta_{mm'}$$

$$\sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell} Y_\ell^{m*}(\theta',\phi')\, Y_\ell^{m}(\theta,\phi) = \delta(\cos\theta-\cos\theta')\,\delta(\phi-\phi').$$

- Any reasonable function $f(\theta,\phi)$ defined for $0\le\theta\le\pi$ and $0\le\phi\le 2\pi$ can be expanded:

$$f(\theta,\phi) = \sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell} A_{\ell m}\, Y_\ell^m(\theta,\phi)$$

$$A_{\ell m} = \int_{-1}^{1} d\cos\theta \int_{0}^{2\pi} d\phi\, Y_\ell^{m*}(\theta,\phi)\, f(\theta,\phi)$$

- Note $\int_{-1}^{1} d\cos\theta \leftrightarrow \int_0^\pi \sin\theta\, d\theta$ and $\sin\theta\, d\theta\, d\phi = d\Omega$.

Spherical harmonics, continued

- The spherical harmonics are designed to interoperate nicely with the other machinery of angular momentum algebra (e.g., ladder operators, rotation matrices, and Clebsch-Gordan coefficients).
- It is possible to bypass ever using the associated Legendre functions by working with $Y_\ell^m(\theta, 0)$ instead. This removes any confusion about the definition of the functions.

Spherical harmonics, basic properties

- $[Y_\ell^m(\theta,\phi)]^* = (-1)^m Y_\ell^{-m}(\theta,\phi)$
- $Y_\ell^m(\pi-\theta,\phi+\pi) = (-1)^\ell Y_\ell^m(\theta,\phi)$ – parity
- $Y_\ell^0(\theta) = \sqrt{\frac{2\ell+1}{2\pi}}\, P_\ell(\cos\theta)$
- $Y_\ell^m(\theta,\phi) = Y_\ell^m(\theta,0)\exp(im\phi)$
- $\int_0^{2\pi} Y_\ell^m(\theta,\phi)\, d\phi = 2\pi\, Y_\ell^0(\theta)\, \delta_{m0}$

Spherical harmonics, addition theorem

- Define $\cos\gamma = \cos\theta_1 \cos\theta_2 + \sin\theta_1 \sin\theta_2 \cos(\phi_1 - \phi_2)$, then

$$P_\ell(\cos\gamma) = \frac{4\pi}{2\ell+1} \sum_{m=-\ell}^{\ell} Y_\ell^{m*}(\theta_1, \phi_1)\, Y_\ell^m(\theta_2, \phi_2).$$

- Using the generating function, we can deduce the expansion

$$\frac{1}{|\vec{\boldsymbol{r}}_1 - \vec{\boldsymbol{r}}_2|} = \sum_{\ell=0}^{\infty} \frac{4\pi}{2\ell+1} \frac{r_<^\ell}{r_>^{\ell+1}} \sum_{m=-\ell}^{\ell} Y_\ell^{m*}(\theta_1, \phi_1)\, Y_\ell^m(\theta_2, \phi_2),$$

where the position vectors $\vec{\boldsymbol{r}}_1$ and $\vec{\boldsymbol{r}}_2$ are described in spherical coordinates by (r_1, θ_1, ϕ_1) and (r_2, θ_2, ϕ_2), respectively, $r_<$ is the lesser of $\{r_1, r_2\}$, and $r_>$ is the greater of $\{r_1, r_2\}$. The angle γ is the angle between $\vec{\boldsymbol{r}}_1$ and $\vec{\boldsymbol{r}}_2$.

Spherical harmonics, reduction of a product

- Reduction of a product of spherical harmonics:

$$Y_{\ell_1}^{m_1}(\theta,\phi)\,Y_{\ell_2}^{m_2}(\theta,\phi) = \sum_{L,M}\sqrt{\frac{(2\ell_1+1)(2\ell_2+1)}{4\pi(2L+1)}} \times \langle \ell_1 m_1 \ell_2 m_2 | LM\rangle \langle \ell_1 0 \ell_2 0 | L0\rangle Y_L^M(\theta,\phi).$$

  The sum is over all (L, M) where the Clebsch-Gordan coefficients are non-zero. Note $M = m_1 + m_2$, so it is really only a sum over L, between $|\ell_1 - \ell_2|$ and $\ell_1 + \ell_2$.

- This formula is useful if you need to integrate a product of three or more spherical harmonics over all θ and ϕ.

Another addition theorem

- Suppose the we have vectors $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$. Then define $\vec{\boldsymbol{c}} = \vec{\boldsymbol{a}} + \vec{\boldsymbol{b}}$, $\vec{\boldsymbol{a}} = a\hat{\boldsymbol{a}}$, $\vec{\boldsymbol{b}} = b\hat{\boldsymbol{b}}$, and $\vec{\boldsymbol{c}} = c\hat{\boldsymbol{c}}$. It is then a fact that

$$c^L\, Y_L^M(\hat{\boldsymbol{c}}) = \sum_{\substack{\ell_1+\ell_2=L \\ m_1+m_2=M}} a^{\ell_1} b^{\ell_2} \langle \ell_1 m_1 \ell_2 m_2 | LM \rangle \times \sqrt{\frac{4\pi(2L+1)!}{(2\ell_1+1)!(2\ell_2+1)!}} Y_{\ell_1}^{m_1}(\hat{\boldsymbol{a}}) Y_{\ell_2}^{m_2}(\hat{\boldsymbol{b}}).$$

- The formula is derived by M. E. Rose, in the Journal of Mathematics and Physics **37**, 215-222 (1958). See also Marcos Moshinsky, Nuclear Physics **13**, 104-116 (1959), Eqs. (46-48), `https://doi.org/10.1016/0029-5582(59)90143-9`.

- It is useful for calculating Van der Waals interactions and in three-body quantum mechanics. I *do not* expect you to do anything with this equation.

Laplace equation in spherical coordinates

- If we are solving $\nabla^2 \Psi = 0$ in a region that includes $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq 2\pi$, the solution must then be

$$\Psi(r, \theta) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \left(A_{\ell m} r^{\ell} + \frac{B_{\ell m}}{r^{\ell+1}} \right) Y_{\ell}^{m}(\theta, \phi),$$

  where $A_{\ell m}$ and $B_{\ell m}$ are constants determined by boundary conditions.

- If our region includes the origin (or $r \to \infty$) and Ψ is required to be finite there, then we must have $B_{\ell m} = 0$ (or $A_{\ell m} = 0$).